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# Reshaping Reasoning in LLMs: A Theoretical Analysis of RL Training Dynamics through Pattern Selection

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## Abstract

1 Reinforcement learning (RL) has become a central driver of stronger reasoning  
2 in language models, yet the quantitative mechanism behind these gains remains  
3 unclear. We study RL training dynamics by pairing empirical measurements with a  
4 theoretical model that makes the comparison unit explicit: method, reward type,  
5 reasoning pattern, token group, and success metric. Across reasoning-pattern-  
6 level and token-level analyses, we find a consistent empirical pattern: individual  
7 reasoning patterns retain relatively stable success rates during training, while RL  
8 concentrates its updates on a sparse subset of critical tokens and shifts probability  
9 mass toward higher-success-rate patterns. This evidence frames the performance  
10 gain as a redistribution over existing reasoning strategies rather than an unexplained  
11 aggregate improvement. Building on this observation, we formalize reasoning as a  
12 two-stage process and analyze two reward settings, verifiable reward (RLVR) and  
13 the model’s internal feedback (RLIF). For RLVR, the theory separates a favorable  
14 regime in which models converge readily to the best-success pattern from a harder  
15 regime in which weaker initial reasoning quality creates an entanglement-stage  
16 optimization barrier. For RLIF, the analysis explains why internal rewards can  
17 produce early performance gains while also identifying how continued optimization  
18 can degrade performance below the base model. Additional experiments and simu-  
19 lations support these empirical regularities and theoretical predictions, clarifying  
20 when RL improves reasoning, which comparisons reveal the improvement, and  
21 why base-model quality materially affects practical outcomes.

## 22 1 Introduction

23 Recent reasoning-focused language models, including Gemini2.5 [8], Qwen3 [40], and DeepSeek-R1  
24 [14], obtain strong results on complex logical benchmarks such as mathematics [30, 44, 41] and  
25 programming [49, 40, 8]. A common ingredient behind these gains is reinforcement learning (RL),  
26 whose practical value is now evident from model-level performance improvements. The central  
27 question is therefore more precise: which measured quantities change during RL training, and how  
28 do those changes translate into better reasoning accuracy?

29 Prior work has begun to answer this question through several quantitative lenses. By comparing  
30 pass@k, Yue et al. [42] show that RL post-training often does not exceed the coverage of the base  
31 model, suggesting that the observed gains may come from selecting among existing reasoning patterns  
32 rather than inventing entirely new ones. From an entropy perspective, Cui et al. [9], Zhang et al. [45]  
33 prove that RL-based methods can reduce policy entropy, and Cui et al. [9] further connect entropy  
34 changes with model performance. At the token level, Wang et al. [33], Huan et al. [17] likewise find  
35 that RL updates concentrate on a sparse subset of critical tokens. These comparisons are informative,  
36 but they leave the full training trajectory under-specified: the field still lacks a unified account of how

37 token-level shifts, pattern-level success rates, and aggregate performance margins interact across RL  
38 training.

39 We address this gap with a systematic empirical analysis that makes each comparison axis explicit.  
40 At the reasoning-pattern level, we use LLM-based and rule-based procedures to extract response  
41 patterns, measure each pattern’s success rate, and track how its distribution changes over training.  
42 At the token level, we compare next-token prediction ranks before and after RL to identify where  
43 the policy actually moves. Relative to previous analyses [17, 42], this paired evidence gives a  
44 sharper quantitative framing than pass@k- and token-only comparisons [17, 42]: RL improves  
45 performance primarily by optimizing a sparse set of critical tokens, thereby *reshaping reasoning*  
46 *pattern distributions* toward patterns that already have higher empirical success rates. Importantly, the  
47 success rates of individual patterns remain comparatively stable, so the dominant measured effect is a  
48 change in pattern selection rather than a broad improvement in every pattern’s conditional accuracy.

49 Guided by these empirical regularities, we develop a theoretical framework that models reasoning as a  
50 two-stage question-reason-answer process  $q \rightarrow r \rightarrow a$ : first selecting a reasoning pattern conditioned  
51 on the question,  $\pi(r|q)$ , and then generating an answer conditioned on both the question and the  
52 selected pattern,  $\pi(a|r, q)$ . This decomposition aligns the theory with the empirical comparisons:  
53 RL can be studied as changing the distribution over  $r$  while the success rate associated with each  
54  $r$  remains comparatively stable. We analyze two representative reward settings, verifiable reward  
55 (RLVR) [14, 30] and the model’s internal feedback (RLIF) [47, 4]. For RLVR, the theory predicts  
56 convergence toward the pattern  $r^*$  with the highest success rate and distinguishes two regimes: strong  
57 base models exhibit rapid movement toward optimal patterns, whereas weaker base models can enter  
58 an entanglement stage that slows or destabilizes optimization. For RLIF, the analysis explains both  
59 the practical early gains from internal rewards and the observed possibility that continued training  
60 converges to a state worse than the base model. Additional experiments and simulations verify these  
61 pattern-level and method-level predictions.

62 The main contributions of this paper are highlighted as follows:

- 63 • We provide a quantitative, pattern-level and token-level account of RL training dynamics. By  
64 comparing next-token rank shifts with LLM-based and rule-based pattern measurements, we show  
65 that RL concentrates updates on a sparse set of critical tokens and shifts probability mass toward  
66 higher-success-rate reasoning patterns. Across training, individual pattern success rates remain  
67 relatively stable, making the redistribution over patterns the key empirical driver of the aggregate  
68 performance change.
- 69 • We introduce a formal two-stage framework for reasoning,  $q \rightarrow r \rightarrow a$ , and use it to analyze two  
70 common RL reward types under matched comparison units. For RLVR, we prove convergence  
71 toward the highest-success reasoning pattern and characterize a favorable rapid-convergence regime  
72 versus an entanglement regime for weaker base models. For RLIF, we theoretically justify early  
73 performance improvements while identifying conditions under which internal-feedback training  
74 can ultimately underperform the base policy.
- 75 • We validate the theoretical predictions with case studies, additional experiments, and numerical  
76 simulations that preserve the same comparison structure across methods, patterns, and metrics.  
77 The resulting evidence links empirical margins to the theory and offers practical guidance for  
78 interpreting and improving RL-based LLM post-training.

## 79 2 Related Works

80 **Reinforcement Learning for LLMs** Reinforcement learning has produced measurable gains for  
81 large language models (LLMs), most prominently in preference alignment [25, 48] and in math or  
82 programming benchmarks where outcomes can be compared directly [30, 20, 22]. The reward signal  
83 is the key comparison variable across these settings. Earlier RL systems typically depended on reward  
84 models trained from human annotations [26, 25, 2], whereas recent reasoning systems increasingly  
85 use lower-cost, easier-to-audit reward sources: verifiable correctness in reinforcement learning with  
86 verifiable rewards (RLVR) [30, 40, 14] and intrinsic model-derived scores in reinforcement learning  
87 with internal feedback (RLIF) [47, 4]. Algorithmic variants such as GRPO [30], DAPO [41], and  
88 VAPO [43] further show that the practical performance of RL depends not only on the reward type  
89 but also on how the training procedure reallocates probability mass across responses.

**Theoretical Analyses and Mechanism Interpretation of RL for LLMs.** A growing literature studies why RL improves LLMs, with evidence reported at several levels of granularity. Early theoretical and algorithmic work emphasized high-level design choices such as reward construction [28, 18, 38] and training algorithms [37, 11, 21]. As RLVR and RLIF made reward definitions simpler and more measurable, later work began to isolate the mechanisms behind the observed accuracy improvements. Shao et al. [29] argue that spurious rewards can unlock latent pretraining skills; Agarwal et al. [4] frame improvements through entropy minimization; and Gandhi et al. [13] identify behavioral correlates such as backtracking. Additional measurements from [46] show that RL fine-tuning amplifies pretrained behaviors and pushes outputs toward dominant formats, while Wang et al. [34], Cui et al. [10] connect this pattern to entropy collapse. Our analysis complements these perspectives by using reasoning patterns as the unit of comparison: we measure pattern distributions and success rates during training, then provide a theoretical account of why RL shifts probability toward higher-success patterns under RLVR and why RLIF can exhibit both early gains and later degradation.

### 3 Preliminaries

**Reinforcement Learning for LLMs** Let  $\pi_\theta$  denote the language-model policy with parameters  $\theta$ . For an input question  $\mathbf{x} = (x_0, x_1, \dots, x_n)$ , the policy samples an answer  $\mathbf{y} = (y_0, y_1, \dots, y_m)$ . We use the following objective to make the quantitative tradeoff explicit: the first term measures expected reward on the data distribution, while the second term penalizes divergence from the reference policy.

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_\theta(\cdot | \mathbf{x})} [r_\phi(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}}[\pi_\theta(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (3.1)$$

where  $\pi_{\text{ref}}$  is the base reference policy, and  $\beta$  controls how strongly the optimized policy is constrained to remain close to that baseline. This form lets us compare methods by holding the policy class fixed while changing the reward signal and the resulting empirical behavior.

The reward function  $r_\phi(\mathbf{x}, \mathbf{y})$  can be instantiated by a trained reward model [25] or by a rule-based correctness score [30]. In this paper, the main comparisons focus on two reward types with different observability: verifiable reward (RLVR) [14, 30], which directly scores answer correctness, and the model’s internal feedback (RLIF) [47, 4], which uses an intrinsic signal derived from the model distribution.

In RLVR,  $r_\phi(\mathbf{x}, \mathbf{y})$  is a binary correctness metric: a response receives reward 1 exactly when the answer  $\mathbf{y}$  matches the ground truth for  $\mathbf{x}$ , and reward 0 otherwise. A typical reward function is defined as:

$$r_\phi(\mathbf{x}, \mathbf{y}) = \begin{cases} 1 & \text{if } \mathbf{y} = \text{the ground truth of } \mathbf{x}, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

Thus, RLVR gives a clean empirical comparison between successful and unsuccessful responses under an externally verifiable metric.

RLIF [47, 4] instead assigns rewards using only model-internal statistics. Following Zhao et al. [47], we take the RLIF reward to be the negative average KL divergence between a uniform distribution  $U$  over the vocabulary  $\mathcal{V}$  and the model’s next-token distribution:

$$r_\phi(\mathbf{x}, \mathbf{y}) = \frac{1}{|\mathbf{y}|} \sum_{i=1}^{|\mathbf{y}|} \mathbb{D}_{\text{KL}}(U \parallel \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})) = -\frac{1}{|\mathbf{y}| \cdot |\mathcal{V}|} \sum_{i=1}^{|\mathbf{y}|} \sum_{j=1}^{|\mathcal{V}|} \log(|\mathcal{V}| \cdot \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})). \quad (3.3)$$

Because this reward is not tied to an external ground truth answer, the relevant quantitative question differs from RLVR: we must compare not only early reward-driven gains but also the final accuracy relative to the base policy.

The rest of the paper uses these two objectives as matched baselines for empirical and theoretical analysis. We track how each reward changes token ranks, reasoning-pattern distributions, pattern-level success rates, and aggregate performance, while preserving the distinction between externally verified correctness and internally generated feedback.

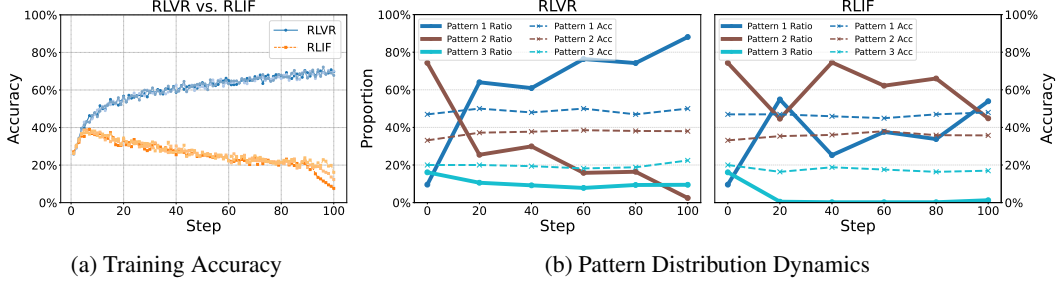


Figure 1: **(a)** Matched training-procedure comparison for RLVR and RLIF on MATH. Across multiple random seeds, RLVR shows steady gains and convergence, whereas RLIF improves early and then declines. **(b)** Pattern-distribution and pattern-accuracy comparison during training. Solid lines report the response share of each pattern among all patterns (left axis), and dotted lines report that pattern’s accuracy (right axis). Patterns are ordered from highest to lowest accuracy, making the main empirical margin clear: RLVR reallocates probability toward higher-accuracy patterns and away from lower-accuracy patterns, while RLIF has less stable pattern dynamics.

## 4 Experimental Exploration for RL Training

To quantify how RL changes reasoning behavior, we evaluate the same base models, datasets, rewards, and training trajectories through three increasingly fine-grained comparisons. First, the training-procedure comparison shows a clear method-level split: RLVR yields steady accuracy gains, whereas RLIF can improve early and then lose performance under continued training. Second, the reasoning-pattern comparison explains this margin by tracking both pattern distributions and pattern-specific success rates; RLVR shifts probability toward higher-success-rate patterns, while RLIF does not reliably specialize in those patterns. Third, the token-level comparison shows that the pattern shift is driven by rank changes in only a small fraction of tokens. Relative to prior work [17, 42], these measurements provide a more explicit quantitative chain from token updates to pattern redistribution to final model performance: RL primarily optimizes a sparse subset of critical tokens, thereby *reshaping reasoning pattern distributions* to affect accuracy.

### 4.1 Training Procedure Overview for RLVR and RLIF

We begin with the aggregate training curves for RLVR and RLIF under a controlled evaluation condition. The base model is Qwen2.5-3B [39], the training dataset is MATH [16], and all settings other than the reward algorithm are held fixed, so the comparison isolates the effect of verifiable versus internal feedback.

Figure 1a makes the contrast visible in the same metric over the same training schedule. RLVR produces a monotone, stable improvement trend that gradually converges, whereas RLIF shows a favorable early margin followed by a decline under longer training; in some runs, the final RLIF model falls below the original base model. This divergence motivates the finer-grained analyses below: the aggregate curve identifies the performance gap, while pattern-level and token-level measurements identify where that gap comes from.

### 4.2 Reasoning-pattern Level Analysis

To explain the RLVR–RLIF performance gap beyond aggregate accuracy, we use reasoning patterns as the comparison unit. We first collect base-model responses and use GPT-4o [19] to group them into categories defined by keywords and logical structure, with a description attached to each pattern. Holding these pattern definitions fixed, we then measure two quantities throughout RL training: the distribution of responses across patterns and the success rate associated with each pattern.

Figure 1b reports this matched analysis for Number Theory responses from MATH [16]; Appendix A repeats the comparison on additional models and datasets. Reading the solid distribution curves together with the dotted success-rate curves yields three consistent empirical findings:

- RLVR-trained models consistently increase the response share of patterns with higher measured success rates, providing a direct explanation for the steady aggregate accuracy gains.
  - RLIF exhibits less stable pattern-distribution dynamics and does not consistently concentrate probability on the higher-success patterns.
  - For both methods, the success rate of a fixed individual pattern remains comparatively stable across training, so the main performance margin comes from redistribution across patterns.
- This analysis localizes the method-level difference: RLVR and RLIF differ less in per-pattern conditional success than in how much probability they allocate to each pattern. We next test the lower-level mechanism that can produce this redistribution.

### 4.3 Token-level Analysis

To connect pattern redistribution to the underlying policy change, we conduct a token-level comparison inspired by Huan et al. [17], Yue et al. [42]. We sample base-model responses for GSM8K [6], MATH [16], and AIME [7], then evaluate the same token sequences under the base and RL-enhanced models. A token’s rank is its probability rank among all vocabulary tokens given the preceding context. Table 1 shows that token ranks are largely preserved: in most dataset-method-step cells, fewer than 10% of tokens experience a rank shift. The practical implication is favorable for the pattern-selection account: RL does not rewrite most of the reasoning trace, but selectively changes a small set of high-leverage tokens that can redirect the model toward different reasoning patterns.

Together, the aggregate, pattern-level, and token-level results provide a consistent quantitative account of RL training dynamics: *RL primarily optimizes a sparse subset of critical tokens, thereby reshaping reasoning pattern distributions to affect model performance. Moreover, the intrinsic success rate of individual patterns remains relatively stable during training.*

These empirical regularities define the quantities that the theory must explain: sparse token-level movement, stable pattern-level success rates, and method-dependent redistribution over patterns. The next section builds a mathematical framework around exactly these measured comparisons.

Table 1: Token ranking-change ratio after RL, comparing identical base-model responses under RLVR and RLIF across datasets and training steps.

| Task   | Method | Step20 | Step40 | Step60 | Step80 | Step100 |
|--------|--------|--------|--------|--------|--------|---------|
| GSM8K  | RLVR   | 5.2%   | 6.0%   | 6.6%   | 7.1%   | 7.3%    |
|        | RLIF   | 6.8%   | 8.0%   | 8.9%   | 9.5%   | 10.1%   |
| MATH   | RLVR   | 5.3%   | 5.8%   | 6.1%   | 6.5%   | 6.6%    |
|        | RLIF   | 6.1%   | 7.0%   | 7.7%   | 8.3%   | 8.8%    |
| AIME24 | RLVR   | 5.1%   | 5.7%   | 5.9%   | 6.1%   | 6.3%    |
|        | RLIF   | 5.6%   | 6.5%   | 7.4%   | 8.2%   | 8.6%    |

## 5 Theoretical Characterization of RL Training Dynamics

The empirical results identify a specific quantitative structure: RL changes the distribution over reasoning patterns more than it changes the success rate of each fixed pattern. This section turns that structure into a theoretical model. We represent reasoning as a two-step process—selecting a reasoning pattern and then deriving an answer—so that the theory compares methods by how they move probability mass across patterns with different success rates. Within this framework, RLVR admits two regimes with sharply different convergence behavior, and RLIF admits a favorable initial effect together with a possible long-run degradation below the base model.

### 5.1 A Theoretical Framework for Reasoning Models

**Abstract Reasoning Process.** We formalize the comparison unit used in the experiments. Given a question  $q$ , the model first selects a reasoning pattern  $r$  from candidate patterns  $\mathcal{R} = \{r_1, r_2, \dots\}$  and then generates a final answer  $a \in \mathcal{A} = \{a_1, a_2, \dots\}$ . Pattern  $r_i$  is selected with probability  $p(r_i|q)$ , and each pattern is associated with its own success rate. This abstraction lets the theory preserve the empirical distinction between a method’s pattern distribution and a pattern’s conditional accuracy. Within this framework, we can reformulate the RLIF optimization objective (Eq. 3.3) as

210 follows<sup>1</sup>:

$$r_\phi(\{\mathbf{r}, \mathbf{a}\}, q) := - \left( \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_\theta(\mathbf{r}_i|q)) + \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_\theta(\mathbf{a}_j|q, \mathbf{r})) \right) \quad (5.1)$$

211 **Policy Parameterization.** Let  $\mathcal{V} = \{\mathbf{q}\} \cup \mathcal{R} \cup \mathcal{A}$  denote the vocabulary set and  $\mathcal{Y} = (r, a)_{r \in \mathcal{R}, a \in \mathcal{A}}$   
 212 represent the set of output sequences. Given a question  $\mathbf{q}$ , the language model produces a distribution  
 213 over output sequences  $\mathbf{y} \in \mathcal{Y}$  autoregressively:

$$\text{(general policy)} \quad \pi_\theta(\mathbf{y}|\mathbf{x}) = \prod_{l=1}^{|\mathbf{y}|} \pi_\theta(\mathbf{y}_l|\mathbf{x}, \mathbf{y}_{<l}) = \prod_{l=1}^{|\mathbf{y}|} \text{softmax}(f_\theta(\mathbf{x}, \mathbf{y}_{<l}))_{\mathbf{y}_l}, \quad (5.2)$$

214 where  $f_\theta : \mathcal{V} \rightarrow \mathbb{R}^{|\mathcal{V}|}$  is parameterized by  $\theta$  and produces the logits used to compare next-token  
 215 probabilities at position  $l$ . The next token follows  $\text{softmax}(\mathbf{z})_v := \exp(\mathbf{z}_v) / \sum_{v' \in \mathcal{V}} \exp(\mathbf{z}_{v'})$  for  
 216  $\mathbf{z} = f_\theta(\mathbf{x}, \mathbf{y}_{<l})$ , matching the token-rank quantities measured in the experimental section.

217 Because practical language models are too complex for clean optimization guarantees [3, 23], we  
 218 follow prior theoretical work [27, 24, 10, 45] and analyze a *tabular policy*. This parameterization is a  
 219 controlled special case of Eq. 5.2: each output receives its own trainable logit for the corresponding  
 220 last token, allowing us to isolate how probability mass moves among reasoning patterns, for  $\theta \in$   
 221  $\mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$ :

$$\text{(tabular policy)} \quad \pi_\theta(\mathbf{y}_l|\mathbf{x}, \mathbf{y}_{<l}) = \pi_\theta(\mathbf{y}_l|\mathbf{y}_{l-1}) = \text{softmax}(\theta_{:, \mathbf{y}_{l-1}})_{\mathbf{y}_l}, \quad (5.3)$$

222 where  $\theta_{:, \mathbf{y}_{l-1}} \in \mathbb{R}^{|\mathcal{V}|}$  is the column of  $\theta$  corresponding to  $\mathbf{y}_{l-1}$ . This simplified policy keeps the  
 223 relevant quantitative objects—pattern probabilities, success rates, and accuracy—explicit.

224 **Optimization for RL.** We analyze training dynamics by combining the tabular policy in Eq.5.3  
 225 with the RL objective  $\phi_{\text{RL}}(\theta)$  in Eq.3.1. In the small-learning-rate limit, the policy-gradient trajectory  
 226 is represented by gradient flow:

$$\frac{d}{dt} \theta(t) = \nabla \phi_{\text{RL}}(\theta(t)), \quad t \geq 0 \quad (5.4)$$

227 where  $\theta(t)$  denotes the policy parameters at training time  $t$ , initialized at the reference policy with  
 228  $\theta(0) = \theta_{\text{ref}}$ .

229 The experiments repeatedly show that success rates for fixed reasoning patterns are relatively stable  
 230 during training. We encode this measured regularity by assuming that RL primarily changes the  
 231 mapping from questions to patterns, while the pattern-to-answer success probability is fixed over the  
 232 time scale analyzed here:

233 **Assumption 5.1.** *The success rate for each reasoning pattern  $\mathbf{r}_i \in \mathcal{R}$  to provide the correct answer*  
 234  *$\mathbf{r}^*$ , defined as  $\pi_\theta(\mathbf{a}^*|\mathbf{q}, \mathbf{r}_i)$ , remains constant during training.*

235 Under Assumption 5.1, we write  $p^*(\mathbf{r}) = \pi_\theta(\mathbf{a}^*|\mathbf{q}, \mathbf{r})$  for the fixed success rate of pattern  $\mathbf{r}$ . We  
 236 first derive the optimal policy for the RL objective (Eq.3.1) under the general autoregressive policy  
 237 (Eq.5.2), and then use the tabular policy (Eq. 5.3) to compare the training dynamics of RLVR and  
 238 RLIF.

## 239 5.2 The Optimal Policy for RL

240 The optimal-policy calculation makes the empirical comparison precise. RLVR rewards a pattern in  
 241 proportion to its success rate, whereas RLIF rewards a confidence-related quantity that need not rank  
 242 patterns by correctness. We therefore characterize the optimizer for both objectives before studying  
 243 the path taken during training:

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<sup>1</sup>For simplicity, we omit the normalizing coefficient  $1/|\mathcal{Y}|$  from Eq. 3.3, which can be treated as a constant  $1/2$  in our framework.

244 **Proposition 5.2.** Suppose we maximize the RL objective (Eq 3.1) using a general autoregressive  
 245 policy (Eq 5.2) and Assumption 5.1 holds. Then, the optimal policy satisfies:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}, \quad (5.5)$$

246 where  $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})$  is the normalizing coefficient, and  $R(\mathbf{r})$  denotes the  
 247 reasoning path reward. Specifically:

- 248 • For RLVR,  $R(\mathbf{r})$  equals the success rate of reasoning pattern  $\mathbf{r}$ , i.e.,  $R_{\text{RLVR}}(\mathbf{r}) = p^*(\mathbf{r})$ .
- 249 • for RLIF,  $R(\mathbf{r})$  equals to the confidence for the reasoning pattern  $\mathbf{r}$  for final answer distribution,  
 250 i.e.,  $R_{\text{RLIF}}(\mathbf{r}) = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}|\mathbf{q}, \mathbf{r}))$ .

251 Proposition 5.2 separates the two reward mechanisms in the same mathematical form. When  $\pi_{\text{opt}}$   
 252 lies in the autoregressive policy class, RLVR increases the probability of a pattern  $\mathbf{r}$  whenever its  
 253 correctness probability  $p^*(\mathbf{r})$  is large enough that  $\exp(\frac{1}{\beta} p^*(\mathbf{r}))/Z > 1$ . Thus the reward ordering  
 254 and the success-rate ordering coincide. RLIF is different:  $R_{\text{RLIF}}(\mathbf{r})$  depends on confidence in the  
 255 final-answer distribution and does not distinguish the correct answer  $\mathbf{a}^*$  from incorrect alternatives.  
 256 As a result, RLVR’s optimal policy is aligned with accuracy improvement, whereas RLIF’s optimal  
 257 policy has no such guarantee relative to the base model.

258 Moreover, since  $\beta$  is typically small (e.g., 0.001) in practice [26, 5, 44], as  $\beta \rightarrow 0$ , we have:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \lim_{\beta \rightarrow 0} \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) = \begin{cases} 1 & \mathbf{r} = \arg \max_{\mathbf{r}} R(\mathbf{r}) \\ 0 & \text{otherwise} \end{cases}, \quad (5.6)$$

259 In this limit, the policy deterministically selects the pattern with the largest  $R(\mathbf{r})$ , independent of  
 260 the reference model’s initial pattern distribution. The remaining question is dynamic rather than  
 261 static: how quickly does training reach the favorable pattern, and under which reward can the selected  
 262 pattern be misaligned with accuracy? We answer this by analyzing RLVR and RLIF under the tabular  
 263 policy in Eq. 5.3.

### 264 5.3 Training Dynamic Analysis for RLVR

265 The experiments show that RLVR’s practical margin depends strongly on base-model quality. We  
 266 therefore track the probability of the *optimal reasoning pattern*, namely the pattern  $\mathbf{r}^*$  with the  
 267 highest success rate for reaching the correct answer  $\mathbf{a}^*$ ,  $\mathbf{r}^* = \arg \max_{\mathbf{r}} p^*(\mathbf{r})$ . Gradient-flow analysis  
 268 yields two regimes. In Regime 1,  $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$  increases steadily until it approaches 1, giving a clean  
 269 theoretical counterpart to the empirical shift toward high-accuracy patterns. In Regime 2, a second-  
 270 best pattern  $\mathbf{r}' \neq \mathbf{r}^*$  temporarily absorbs probability mass and slows the movement toward  $\mathbf{r}^*$ ; after  
 271 this entanglement stage, the dynamics transition to Regime 1 and converge to  $\mathbf{r}^*$ .

272 **Theorem 5.3** (Regime 1: Sufficient Condition for Efficient Convergence). Consider the RLVR (Eq  
 273 3.2) with  $\beta = 0$  for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1  
 274 holds. Let  $\mathbf{r}^*$  be the optimal reasoning pattern, if the overall accuracy of the initialized model  $\pi_{\theta_{\text{ref}}}$ ,  
 275 defined as  $\text{ACC}_{\theta_{\text{ref}}} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$ , satisfies:

$$(\text{Regime 1}) \quad \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*, \quad (5.7)$$

276 then for any  $\epsilon > 0$ , there exists  $T_1 = \mathcal{O}(\frac{1}{\epsilon})$  such that for  $t > T_1$ , we have  $1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) < \epsilon$ .

277 Theorem 5.3 formalizes the favorable case: the base reference model’s aggregate accuracy already  
 278 exceeds the success rate of every non-optimal pattern (Eq 5.7). Under this condition, RLVR reaches  
 279 the optimal pattern efficiently, with convergence time  $\mathcal{O}(1/\epsilon)$ , so the final aggregate accuracy  
 280 approaches the success rate of the best pattern.

281 The second regime captures the less favorable empirical setting in which RLVR is harder to optimize  
 282 [44, 36], typically because the base reference model assigns insufficient probability to the best pattern.  
 283 We define the second-highest-success pattern as  $\mathbf{r}' = \arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$  and analyze how this  
 284 suboptimal pattern can create an entanglement stage that delays optimization toward  $\mathbf{r}^*$ :

**Theorem 5.4** (Regime 2: Slow Convergence for optimal reasoning pattern). *Consider the RLVR (Eq 3.2) with  $\beta = 0$  for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let  $\mathbf{r}^*$  and  $\mathbf{r}'$  be the optimal and second optimal reasoning patterns, if the overall accuracy of the initialized model satisfies:*

$$(Regime\ 2) \quad p^*(\mathbf{r}') > \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}, \quad (5.8)$$

then there exists:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|q)} \left( (C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right), \text{ where } \gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|q)}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|q)} \quad (5.9)$$

with constants  $C_1, C_2$  depending on the success rates of reasoning patterns, such that for we can guarantee the model transform from regime 2 (Eq 5.8) to regime 1 (Eq 5.7), i.e.,  $\text{ACC}_{\theta(t)} > p^*(\mathbf{r}), \forall \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*$  for  $t \geq T_0$ .

Theorem 5.4 studies a clean version of the difficult case: only the optimal pattern and the second-best pattern have success rates above the initialized model’s aggregate accuracy. This setting isolates the measured comparison that matters—whether probability mass moves to the best above-average pattern or remains entangled with a suboptimal above-average pattern. Compared with the favorable condition in Theorem 5.3, more general cases with *at least* two above-average patterns should follow the same intuition, although the proof would require more complex bookkeeping.

The quantitative implication of Theorem 5.4 is the gap between  $T_1$  and  $T_0$ . In Regime 1, Theorem 5.3 gives a polynomial convergence time  $T_1$  in  $1/\epsilon$ . In Regime 2, the transition time  $T_0$  in Theorem 5.4, needed to make aggregate accuracy exceed the suboptimal pattern’s success rate, can grow super-exponentially with  $\gamma_{\pi_{\text{ref}}}$ , the ratio between the total reference probability over  $\mathcal{R}/\{\mathbf{r}'\}$  and the reference probability on  $\mathbf{r}^*$ . Thus, when the base model initially assigns very little probability to the optimal pattern,  $\gamma_{\pi_{\text{ref}}}$  can be large and  $T_0$  can become prohibitively long. This is the theoretical form of the entanglement stage observed in difficult RLVR training.

Sec 6 provides case studies that illustrate these two regimes with concrete training trajectories.

## 5.4 Theoretical Explanation for RLIF

Proposition 5.2 shows why RLIF is not automatically aligned with accuracy: its reward depends on confidence in the final-answer distribution and does not distinguish  $\mathbf{a}^*$  from incorrect answers. This explains the empirical risk that RLIF can finish below the base model. At the same time, prior work reports that internal-feedback training can improve performance without external rewards Agarwal et al. [4], Zhao et al. [47]. We therefore analyze RLIF’s initial derivative to explain how early gains and late degradation can coexist.

We begin with a base model whose answer distribution satisfies the following majority condition:

**Assumption 5.5.** *For the base model, the correct answer  $\mathbf{a}^*$  has the highest probability across all possible answers, i.e.,  $\arg \max_{\mathbf{a} \in \mathcal{A}} \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\text{base}}(\mathbf{a}|q, \mathbf{r}) \pi_{\text{base}}(\mathbf{r}|q) = \mathbf{a}^*$ .*

Because majority voting [35] often improves model performance, strong LLMs commonly place more total probability on the correct answer  $\mathbf{a}^*$  than on any incorrect answer, making Assumption 5.5 plausible for modern models. Under this condition, we study a tractable case with  $|\mathcal{A}| = 2$ , uniformly distributed reasoning-path success rates, and high-entropy pattern selection<sup>2</sup>. We then compute the aggregate-accuracy derivative at initialization:

**Theorem 5.6** (RLIF Increases Overall Accuracy at Initialization). *Consider the RLIF (Eq 5.1) with  $\beta = 0$  for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 and 5.5 hold,  $\pi_{\theta_{\text{base}}}(\mathbf{r}|q) = \frac{1}{|\mathcal{R}|}$  for all  $\mathbf{r} \in \mathcal{R}$ ,  $|\mathcal{A}| = 2$  and success rate for each reasoning path follows a uniform distribution  $p^*(\mathbf{r}) \sim U[0, 1]$ . Then when  $|\mathcal{R}| \rightarrow +\infty$ , the following holds:*

1. The accuracy derivative at  $t = 0$  is positive with probability 1, i.e.  $P(\frac{d}{dt} \text{ACC}_{\theta(t)}|_{t=0} > 0) = 1$ .
2. With probability  $p = 0.5$ , that  $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$ .

<sup>2</sup>Previous work [33] reveals that RL primarily optimizes high-entropy tokens. Here, we consider the highest entropy case:  $\pi_{\theta_{\text{base}}}(\mathbf{r}|q) = 1/|\mathcal{R}|$  for all  $\mathbf{r} \in \mathcal{R}$ .

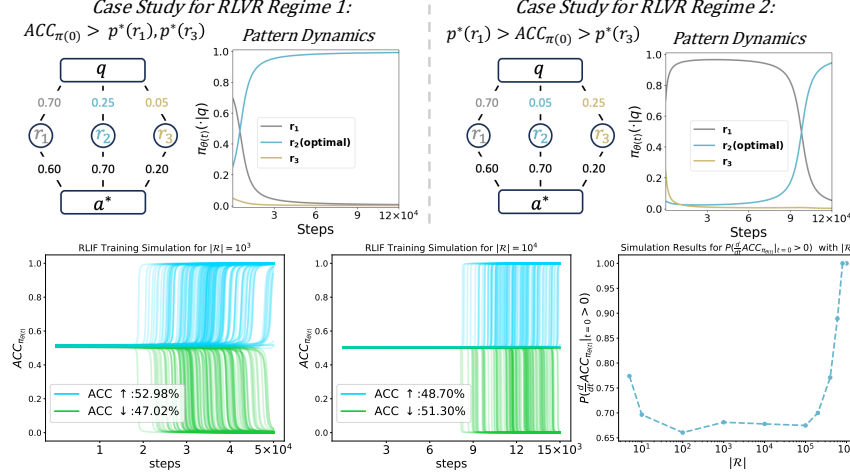


Figure 2: Case studies and simulations aligned with the theoretical comparison metrics. Upper panels: RLVR exhibits rapid convergence when the initialized model is already in Regime 1 (left), but enters a long entanglement phase before convergence in Regime 2 (right). Bottom panels: RLIF simulations report the probability of converging to improved versus degraded final accuracy states (left) and the probability of an initial accuracy gain (right) as  $|R|$  varies.

Theorem 5.6 gives a two-part quantitative explanation of the RLIF curve. The positive derivative at  $t = 0$  explains the early accuracy gain observed in training. But when  $\beta = 0$ , Proposition 5.2 implies deterministic convergence to the pattern maximizing  $R_{\text{RLIF}}(\mathbf{r})$ . The second part of Theorem 5.6 shows that, with probability 0.5, this reward-maximizing pattern is exactly the least accurate pattern:  $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$ . Thus the same internal signal can produce an initial improvement and still have a 50% chance, in this stylized setting, of converging to the lowest-success reasoning path. This matches the empirical pattern in which RLIF starts favorably but can ultimately underperform the base model.

## 6 Case Studies and Numerical Simulations

We use case studies and simulations to check the theory against the same quantities emphasized in the experiments: convergence regime, pattern probability, aggregate accuracy, and final-state comparison. For RLVR, Fig. 2 (upper) contrasts the two regimes in Theorems 5.3 and 5.4. For RLIF, Fig. 2 (bottom) reports repeated simulations in which each reasoning path’s success rate is uniformly sampled, allowing us to compare the probability of initial improvement with the probability of converging to an improved or degraded final state.

### 6.1 Case Studies for RLVR

Section 5.3 predicts two RLVR regimes with different practical time scales. In Regime 1 (Eq 5.7), the probability of the optimal pattern  $\pi_{\theta}(\mathbf{r}^*|q)$  increases monotonically until convergence. In Regime 2 (Eq 5.8), probability mass is initially entangled with a strong but suboptimal pattern, so the model needs a much longer transition period before it enters the rapid-convergence behavior of Regime 1. The two case studies make this margin visible:

- Fig. 2 (upper left) shows the favorable condition in Eq 5.7: initialized aggregate accuracy exceeds every non-optimal pattern’s success rate, and the model rapidly concentrates on the optimal reasoning pattern.
- Fig. 2 (upper right) illustrates Theorem 5.4. When the suboptimal pattern  $\mathbf{r}'$  has a success rate above the initialized aggregate accuracy, training first enters an entanglement phase. The transition time  $T_0$  from Theorem 5.4 becomes large when  $\gamma_{\pi_{\text{ref}}}$  in Eq 5.9 is large; here  $\gamma_{\pi_{\text{ref}}} = 6$  because  $(0.25 + 0.05)/0.05 = 6$ . Since  $T_0$  scales with  $\gamma_{\pi_{\text{ref}}}^{\gamma_{\pi_{\text{ref}}}}$ , convergence to the optimal pattern is substantially delayed, and the pattern dynamics in Fig. 2 make this prolonged entanglement visible.

## 6.2 Numerical Simulations For RLIF

Section 5.4 predicts a distinctive quantitative pattern for RLIF under Assumption 5.5 and the distributional conditions in Theorem 5.6: as  $|\mathcal{R}| \rightarrow \infty$ , the initial accuracy derivative is positive with probability 1, yet convergence to the least accurate reasoning path occurs with probability 50%. The simulations in Fig. 2 confirm both sides of this claim. The bottom-left panel shows that the probability of ending in an improved state ( $ACC \uparrow$ ) is approximately balanced with the probability of ending in a degraded state ( $ACC \downarrow$ ), matching the 50% final-state result in Theorem 5.6. Fig. 2 (bottom right) shows that the probability of an initial performance increase approaches 1 for large  $|\mathcal{R}|$ , matching the initialization result in Theorem 5.6. Together, these simulations explain why RLIF can look beneficial early while remaining risky as a long-run optimization method.

## 7 Conclusions and Limitations

This work reframes RL training for reasoning models around the quantitative objects that actually move during optimization. Across empirical analyses, RL primarily changes a sparse set of token ranks and reallocates probability mass across reasoning patterns; the success rate of each fixed pattern remains comparatively stable. This comparison structure explains why RLVR delivers steady gains by concentrating on higher-success-rate patterns, why its convergence depends on base-model quality, and why RLIF can produce an early accuracy improvement while still converging to a degraded final state. The theoretical framework formalizes these observations for RLVR and RLIF, and the case studies and simulations support the predicted regimes, transition delays, and convergence probabilities.

The analysis also has limitations. The reasoning patterns used in the empirical comparisons are identified by LLM-based and rule-based procedures, so their interpretability and classification reliability deserve further validation. The theory intentionally simplifies real model behavior through a pattern-selection abstraction and a tabular-policy analysis; extending the same quantitative comparisons to richer policy classes and more complex reasoning tasks is an important next step. Finally, several conclusions depend on base-model assumptions, so future work should measure how these regimes vary across broader model families, datasets, and reward designs.

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## A Additional Experiments

### A.1 Pattern Distribution Dynamics on RL-enhanced Qwen2.5-3B

Section 4.2 uses MATH Number Theory tasks to measure reasoning-pattern dynamics. Here we repeat the same dynamic comparison on Geometry tasks (Figure 3). The empirical pattern is consistent across domains: under RLVR, response mass gradually shifts toward the higher-accuracy pattern, whereas RLIF produces a less stable pattern-selection trajectory. This supplemental result supports both the main empirical claim and the theoretical distinction between RLVR and RLIF.

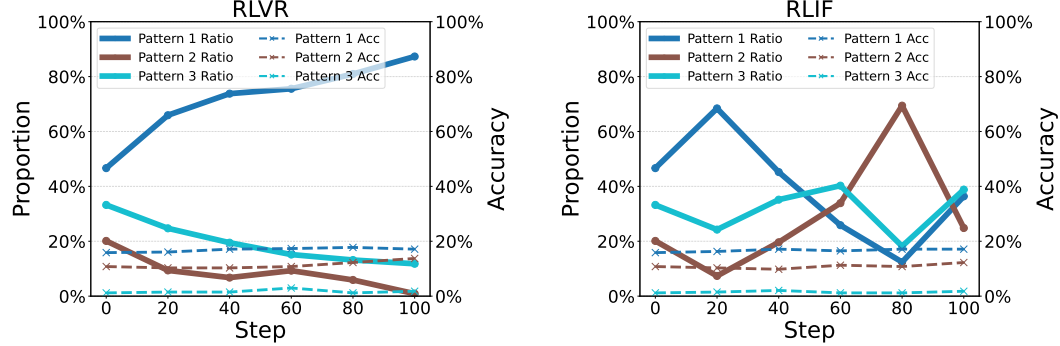


Figure 3: Geometry pattern-distribution dynamics and pattern accuracies for RL-enhanced models. RLVR increasingly selects the highest-accuracy pattern (Pattern 1), whereas RLIF shows less stable selection across patterns.

### A.2 Qwen-2.5-7B-Instruct vs. Qwen-2.5-7B-SimpleRL-Zoo

To test whether the pattern-selection finding generalizes beyond the main model, we compare Qwen-2.5-7B-Instruct [39] with Qwen-2.5-7B-SimpleRL-Zoo [44], an open-source RLVR-enhanced model for mathematical reasoning (Figure 4). The comparison spans number theory, geometry, algebra, calculus, and counting and probability tasks from MATH [16], plus a more complex AMC23 problem [1]. Across these task-level comparisons, the RLVR-enhanced model uses high-accuracy reasoning patterns more frequently and low-accuracy patterns less frequently, matching the main paper’s empirical and theoretical conclusions. Detailed task-specific settings are provided in Appendix B.3.

Across these tasks, the highest-success patterns consistently become more prevalent after RLVR enhancement, reinforcing the comparison reported in Section 4 and the theoretical account in Section 5. Algebra and AMC23-19 in Figure 4 are especially informative because individual pattern success rates remain stable while the pattern distribution changes, providing direct empirical support for Assumption 5.1.

### A.3 Qwen-2.5-32B-Instruct vs. QwQ-32B

We also evaluate whether the pattern-selection account holds at a larger scale by comparing Qwen-2.5-32B-Instruct [39] with QwQ-32B [32], an RLVR-enhanced model built from that base. The evaluation uses four tasks: number theory and geometry from MATH [16], one task from AIME 2024 [7], and one task from OlympiadBench [15]. The key metric is the shift in reasoning-pattern distribution between the base and RLVR-enhanced models.

The resulting distributions show that most QwQ-32B responses align with the patterns that are most accurate for Qwen-2.5-32B, extending the main result to larger models. We omit QwQ pattern-accuracy dots because the API setting does not expose the CoT data needed to let the base model continue generation from a fixed partial reasoning trace. Consequently, we can measure model-level outputs and pattern distributions but not the conditional accuracy of each QwQ reasoning pattern under the continuation protocol described in the “Accuracy Analysis” of Appendix B.2.

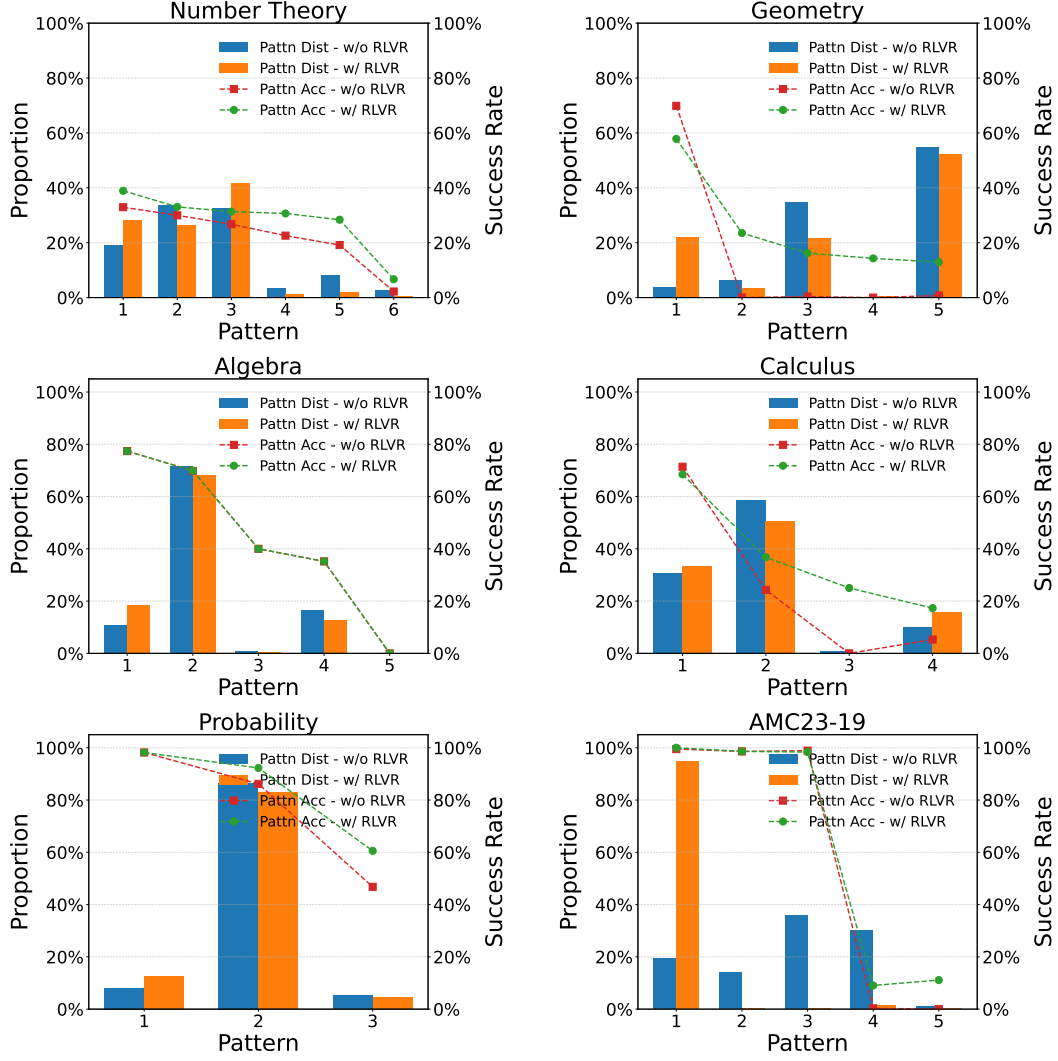
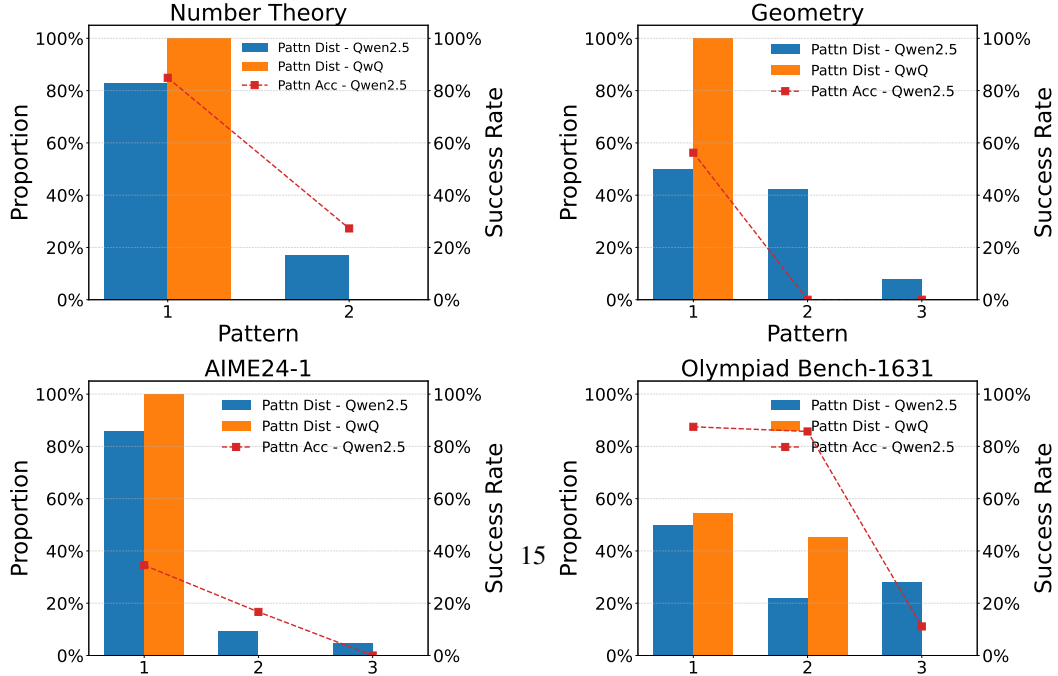


Figure 4: Additional task comparisons between models with and without RLVR enhancement. Bars report each pattern’s response share (Pattern Dist), and dots report the corresponding pattern accuracy or success rate (Pattern Acc). Across tasks, RLVR increases the use of higher-success patterns and reduces the use of lower-success patterns, consistent with the main empirical and theoretical analysis.



## B Experimental Details

### B.1 Experimental Setup

**Entire Training Procedures:** We use Qwen2.5-3B as the shared base model and train on MATH with both RLVR and RLIF. The verl [31] framework provides the training setup, and all parameters match the MATH example in verl except for the reward calculation used by RLIF. We run three random seeds for each method, so the training-curve comparison reflects reward-type differences rather than changes in model, dataset, or hyperparameter settings.

**Token-level Analysis:** We compare Qwen2.5-3B with the corresponding RL-enhanced models on identical token sequences. For GSM8K and MATH, we randomly sample 400 test questions from each dataset; for AIME24, we use the full dataset. For every question, we first sample the base-model answer, concatenate that answer with the original question, and feed the resulting prompt to the RL-enhanced model. Using the OpenAI library interface, we compute each token’s probability and its rank among all vocabulary-token probabilities at the same position. The reported metric is the proportion of tokens whose rank changes after RL, which directly measures how sparse the token-level update is.

**Reasoning-pattern Level Analysis:** We use two complementary pattern-level comparisons. First, for Qwen2.5-3B and its RL-enhanced checkpoints, we evaluate every 20 steps to measure the dynamics of pattern distributions during training. Second, to test generality across models and scales, we compare Qwen-2.5-7B-SimpleRL-Zoo and QwQ-32B with their corresponding base models. For these released models, intermediate checkpoints are unavailable, so the comparison is before-versus-after RL rather than a full training trajectory.

### B.2 Reasoning Patterns Analysis Pipeline

**Model Inference:** We sample questions across task types such as Geometry and Algebra and across difficulty levels including Level 4 and Level 5 from datasets such as MATH. For pattern-distribution dynamics, the model generates 1024 answers per question. For before-versus-after RL comparisons, the number of sampled answers ranges from 64 to 1024 depending on the task. These response sets define the empirical distribution used for subsequent pattern extraction and classification.

**Reasoning Patterns Extraction:** We sample a subset of base-model responses and use the GPT-4o [19] API to summarize pattern categories from those samples. The full extraction prompt is provided in Appendix D.1. We set the GPT-4o temperature to 0 so that pattern definitions are deterministic for a fixed response subset.

**Responses Classification:** We then use GPT-4o to classify every response into the fixed pattern categories, again with temperature 0. The full classification prompt is provided in Appendix D.2. This step yields the pattern-distribution metric used throughout the empirical comparisons.

**Accuracy Analysis:** To isolate the contribution of pattern distribution from changes in the reasoning trace itself, we estimate pattern accuracy by fixing a partial reasoning process. We concatenate the original question with a partial base-model response that represents a specific pattern, remove the sentence containing the final answer, and append the preceding sentence to the question. The RL-enhanced model then continues from this fixed prompt and predicts the answer. We judge the predicted answer and use that outcome as the estimated success rate of the corresponding reasoning pattern. This protocol makes the reported pattern-accuracy values conditional on a fixed pattern rather than on the model’s changing pattern-selection frequency.

### B.3 Details for Each Task

We next describe each evaluation task, the sample construction, and examples of the reasoning-pattern categories used for comparison (Table 2). Example task prompts are provided in Appendix D.3.

**Number Theory:** This task asks models to solve coin-distribution problems across multiple bags. Bags initially contain equal numbers of coins; after additional coins are received and redistributed

| Task                     | Reasoning Patterns Examples                                     | Common Elements / Key Words                                                         |
|--------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Number Theory            | Modular Congruence with Coefficient Simplification              | Uses modular arithmetic to express divisibility conditions.                         |
|                          | Inequality-Driven Search for Minimal Solution                   | Often involves substituting back to compute the total coins.                        |
| Geometry                 | Systematic Inequality Application                               | Explicitly lists and solves each of the three inequalities.                         |
|                          | Verification of Scalene Condition                               | Adding an extra layer of validation to verifies the third side length.              |
| Algebra                  | Iterative Floor Division with Leftover Tracking.                | Uses floor division to compute new cans per step.                                   |
|                          | Recursive Recycling with Aggregated Leftovers.                  | Combines leftovers with newly produced cans before recycling.                       |
| Calculus                 | Direct Simplification and Principal Value Matching              | Assumes inputs fall within the principal range.                                     |
|                          | Interval Analysis with Case Splitting                           | Splits the whole domain into different intervals.                                   |
| Counting and Probability | Direct Probability Setup and Quadratic Solution                 | Explicitly calculates combinations for total and favorable outcomes.                |
|                          | Early Simplification and Cross-Multiplication                   | Early cross-multiplication to eliminate denominators.                               |
| AMC23-Q19                | Prime Factorization and Simplification                          | Counting Digits in Large Numbers.                                                   |
|                          | Rewriting $(8^5)$ as $(2^{15})$ $(15^5)$ as $(3^5 \cdot 5^5)$ . | Final step of counting digits: $(3 \text{ (from 243)} + 15 \text{ (zeros)} = 18)$ . |
| AIME24-Q1                | Direct Equation Setup and Elimination                           | Straightforward and relies on algebraic manipulation.                               |
|                          | Alternative Equation Formulation and Solving                    | Expressing variables in terms of others early on.                                   |
| OlympiadBench-Q1631      | Direct Calculation and Empirical Testing                        | Relying on direct computation and empirical verification.                           |
|                          | Factorization-Based Reasoning                                   | Using algebraic factorization to argue certain terms $y_n$ .                        |

Table 2: Examples of reasoning-pattern categories and identifying elements used to compare pattern distributions and pattern-level success rates across tasks.

equally, the total must exceed a specified value while equality across bags is preserved. The model must output the minimum initial coin count per bag. We derive the task from a level 5 MATH problem (id: test/number\_theory/1055) [16] and create 32 evaluation variants by changing parameters such as bag count and coin totals. We use one-shot prompting with a simple unrelated example to enforce "boxed" answer formatting without steering the reasoning pattern. For pattern extraction, we choose 20 evaluation questions and sample 4 responses each; for distribution comparison, we sample 64 responses per question and classify them into the extracted patterns before and after RL.

**Geometry:** The geometry task gives the model the lengths of two sides of a scalene triangle and asks how many integer-centimeter values are possible for the third side. The task is based on a level 4 MATH geometry problem (id: test/geometry/1046). We use the same evaluation pipeline as in Number Theory: construct 32 synthetic questions, sample 20 questions with 4 base-model responses each from Qwen2.5-7b-Instruct for pattern extraction, and then compare models with and without RLVR on all evaluation questions using 64 responses per question for pattern analysis.

**Algebra:** The algebra task is a can-recycling problem. Starting from a fixed number of old cans, the rule specifies that  $n$  old cans can be recycled into one new can, and the model must compute the total number of cans that can eventually be produced. The quantitative difficulty comes from leftovers: at each iteration, some cans may not divide evenly by  $n$ , but those leftovers can later combine with newly produced cans for additional recycling. The template is a level 4 MATH algebra problem (id: test/algebra/2768). For this and the following supplementary tasks, we keep the dataset size, number of samples, and comparison protocol aligned with the previous tasks.

634 **Calculus:** The calculus task presents a single-variable trigonometric equation together with the  
635 variable’s domain and asks for the number of solutions. Although the task is relatively direct, it  
636 provides a controlled setting for comparing whether RLVR changes the frequency of solution-counting  
637 patterns. The prototype is a level 4 MATH pre-calculus problem (id: test/precalculus/1140).

638 **Counting and Probability:** The counting-and-probability task involves white and black balls.  
639 Two balls are sampled at random, the probability of drawing one black and one white ball is given,  
640 and the number of balls of one color is specified. The model must compute the minimum possible  
641 number of balls of the other color. This level 4 MATH counting-and-probability problem (id:  
642 test/counting\_and\_probability/79) lets us compare algebraic probability setups and simplification  
643 patterns under the same sampling protocol.

644 **AMC23 Question:** We use question 19 from AMC23 [1]. For this task, we randomly sample 48  
645 responses for reasoning-pattern extraction and 1024 responses for the pattern-distribution analysis.

#### 646 B.4 Details for Experiments on QwQ-32B and Qwen-2.5-32B

647 We evaluate QwQ-32B and Qwen-2.5-32B on four tasks. Because API access limits the evaluation  
648 protocol, we randomly select one question per task, sample 64 responses from each model, and  
649 perform reasoning-pattern extraction and classification directly on those responses. For the MATH  
650 tasks, we use number theory question 1055 and geometry question 1046. For the other two tasks, we  
651 use problem 1 from AIME24 and problem 1631 from Olympiad Bench. Table 2 shows representative  
652 pattern categories used in these comparisons.

### 653 C Deferred Proofs

654 This appendix provides the deferred proofs for Proposition 5.2, the first RLVR convergence theorem,  
655 Theorem 5.4, and Theorem 5.6. Each proof keeps the same quantitative comparison units used in the  
656 main text: pattern probability, pattern success rate, aggregate accuracy, and convergence time.

#### 657 C.1 Proof of Proposition 5.2

658 *Proof of Proposition 5.2.* We follow the proof technique of [26] and begin from the KL-regularized  
659 RL objective:

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_{\theta}(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.1})$$

660 Under the two-stage reasoning framework in Section 5.1, the policy first samples a pattern  $\mathbf{r}_i \in \mathcal{R}$   
661 from  $\pi_{\theta}(\mathbf{r}_i|\mathbf{q})$  and then samples an answer  $\mathbf{a} \in \mathcal{A}$  from  $\pi_{\theta}(\mathbf{a}|\mathbf{r}_i)$ . This decomposition rewrites the  
662 objective as a reward comparison over reasoning patterns:

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\mathbf{y}|\mathbf{q})} \left[ \sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q}) \right] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.2})$$

663 Let  $R(\mathbf{r}) = \sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q})$  denote the pattern-level reward. Then:

$$\begin{aligned} \phi_{\text{RL}}(\theta) &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q}), \mathbf{a} \sim \pi_{\theta}(\cdot|\mathbf{r})} \left[ R(\mathbf{r}) - \beta \ln \left( \frac{\pi_{\theta}(\mathbf{r}|\mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a}|\mathbf{r})}{\pi_{\text{ref}}(\mathbf{r}|\mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a}|\mathbf{r})} \right) \right] \\ &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} \left[ R(\mathbf{r}) - \beta \ln \left( \frac{\pi_{\theta}(\mathbf{r}|\mathbf{q})}{\pi_{\text{ref}}(\mathbf{r}|\mathbf{q})} \right) \right] \\ &= -\beta \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} \left[ \ln \left( \frac{\pi_{\theta}(\mathbf{r}|\mathbf{q})}{\frac{1}{Z} \exp \left( \frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\text{ref}}(\mathbf{r}|\mathbf{q})} \right) - \ln Z \right] \\ &= -\beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r}|\mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] + \beta \ln Z. \end{aligned}$$

664 The third equality uses Assumption 5.1. Here,  $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp \left( \frac{1}{\beta} p^*(\mathbf{r}) \right) \pi_{\theta, \text{ref}}(\mathbf{r}|\mathbf{q})$  is the partition  
665 constant that ensures

$$\pi^*(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp \left( \frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\theta}(\mathbf{r}|\mathbf{q}),$$

666 is a valid probability distribution such that  $\sum_{\mathbf{r} \in \mathcal{R}} \pi^*(\mathbf{r}|\mathbf{q}) = 1$ . Since  $Z$  is not a function of  $\mathbf{r}$ .

667 Thus, maximizing Equation C.2 is equivalent to the following KL minimization:

$$\begin{aligned} & \max_{\pi} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] \\ & = \min_{\pi} \beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r}|\mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] - \beta \ln Z. \end{aligned}$$

668 By the nonnegativity and minimizer of KL divergence, the optimal policy for this KL-constrained  
669 reward objective satisfies:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}.$$

670 For RLVR, this pattern reward reduces to the pattern success rate  $R_{\text{RLVR}}(\mathbf{r}) = p^*$ . For RLIF, the  
671 reward is:

$$\begin{aligned} R'(\mathbf{r}) &= - \left( \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_{\theta}(\mathbf{r}_i|q)) + \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_{\theta}(\mathbf{a}_j|q, \mathbf{r})) \right) \\ &= - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|q, \mathbf{r})) - \underbrace{\left( \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_{\theta}(\mathbf{r}_i|q)) + \log |\mathcal{A}| \right)}_c, \end{aligned}$$

672 where the second term is the same for all  $\mathbf{r} \in |\mathcal{R}|$ , for RLIF, the optimal policy can be written as

$$\begin{aligned} \pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) &= \frac{1}{Z} \exp\left(\frac{1}{\beta} R'(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \\ &= \frac{\exp\left(\frac{1}{\beta} R'(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp\left(\frac{1}{\beta} (R'(\mathbf{r}_i))\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}_i|\mathbf{q})} \\ &= \frac{\exp\left(\frac{1}{\beta} (R'(\mathbf{r}) - c)\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp\left(\frac{1}{\beta} (R'(\mathbf{r}_i) - c)\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}_i|\mathbf{q})} \end{aligned}$$

673 Therefore, the RLIF pattern-level reward can be written as  $R_{\text{RLIF}}(\mathbf{r}) = R'(\mathbf{r}) - c =$   
674  $-\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|q, \mathbf{r}))$ . This concludes the proof of the theorem.  $\square$

## 675 C.2 Dynamics of Tabular Policy

676 Consider the RL objective (Eq 3.1), using a tabular policy (Eq 5.3) with Assumption 5.1 holds, the  
677 gradient dynamics of  $\theta$  can be computed as

$$\begin{aligned} \frac{d}{dt} \theta(t) &= \nabla \phi_{\text{RL}}(\theta(t)) \\ &= \nabla \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q})] \\ &= \sum_{\mathbf{r} \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q}) \nabla \pi_{\theta(t)}(\mathbf{r}|\mathbf{q})], \end{aligned}$$

678 Where  $\pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) = \text{softmax}(\theta_{:,q})_{\mathbf{r}}$ ,  $\theta_{:,q} \in \mathbb{R}^{|\mathcal{V}|}$  is the column of  $\theta$  corresponding to  $\mathbf{q}$ , and  
679  $r_{\phi}(\mathbf{r}|\mathbf{q})^{KL}$  refers to the reward with KL divergence, so

$$\begin{aligned} \frac{\partial \theta_{\mathbf{r}_i, \mathbf{q}}(t)}{\partial t} &= \sum_{\mathbf{r}_j \in \mathcal{R}} \left[ r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (\nabla \text{softmax}(\theta_{:,q})_{\mathbf{r}_j})_{\mathbf{r}_i} \right] \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} \left[ r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (-\pi_{\theta(t)}(\mathbf{r}_j|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})) \right] \\ &\quad + r_{\phi}^{KL}(\mathbf{r}_i|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \end{aligned}$$

where the overall accuracy is defined as  $\text{ACC}_{\theta(t)} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$ . For  $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$ , we have

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) = \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)} \cdot \frac{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)}{\partial t}$$

### 682 C.3 Proof of Theorem 5.3

To prove Theorem 5.3, we first show that aggregate accuracy is nondecreasing,  $\text{ACC}_{\theta(t)} \geq \text{ACC}_{\theta(0)}$  for  $t \geq 0$  (Eq C.4). We then lower-bound the optimal pattern probability  $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})$  (Eq C.5) and use this bound to derive the convergence time in Theorem 5.3.

*Proof of Theorem 5.3.* Setting  $\beta = 0$ , Eq C.3 becomes:

$$\frac{\partial \theta_{\mathbf{r}_i, \mathbf{q}}(t)}{\partial t} = \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \cdot (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}). \quad (\text{C.3})$$

For  $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$ , we derive:

$$\begin{aligned} \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)} \cdot \frac{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)}{\partial t} \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}), \end{aligned}$$

and for the accuracy  $\text{ACC}_{\theta(t)}$ :

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_i, \mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} \text{ACC}_{\theta(t)} \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \\ &\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) p^*(\mathbf{r}_i) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)})^2 \geq 0. \quad (\text{C.4}) \end{aligned}$$

As  $\frac{d}{dt} \text{ACC}_{\theta(t)} \geq 0$ , so  $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$  holds for  $t \geq 0$ , for the optimal reasoning pattern  $\mathbf{r}^*$ , let  $\mathbf{r}'$  the suboptimal reasoning pattern with the second highest success rate, defined as  $\mathbf{r}' = \arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$ , we have:

$$\begin{aligned} \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \\ &\geq (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}^*)) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \\ &= (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\geq (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \geq 0. \quad (\text{C.5}) \end{aligned}$$

692 As  $\frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq 0$ ,  $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})$  holds for  $t \geq 0$ , define  $C =$   
 693  $(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(0)}^2(\mathbf{r}^*|\mathbf{q})$ :

$$\frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq C (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2.$$

694 This differential inequality yields:

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) > 1 - \frac{1}{Ct + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})}}.$$

695 For any  $\epsilon > 0$ , there exists  $T_1 = \frac{1}{C} \left( \frac{1}{\epsilon} - \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})} \right) = \mathcal{O}(\frac{1}{\epsilon})$ , such that

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) > 1 - \epsilon.$$

696 This concludes the proof of the theorem.  $\square$

#### 697 C.4 Proof of Theorem 5.4

698 To prove Theorem 5.4, we first upper-bound the probability of the second-best pattern  $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$   
 699 (Eq C.6). We then analyze the probability ratio  $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$  (Eq C.7) and show that after  
 700  $t > T_0$ , aggregate accuracy satisfies  $\text{ACC}_{\theta(t)} \geq p^*(\mathbf{r}')$ .

701 *Proof of Theorem 5.4.* Let  $\mathbf{r}^*$  and  $\mathbf{r}'$  be the optimal and second optimal reasoning patterns. For  
 702  $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$ :

$$\begin{aligned} \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\leq \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}') - p^*(\mathbf{r}')) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\leq \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 + p^*(\mathbf{r}') \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \\ &\leq 2 (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2. \end{aligned}$$

703 This yields:

$$\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \leq 1 - \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}. \quad (\text{C.6})$$

704 Next, define  $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$ , then for all  $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}$ , we have

$$\begin{aligned} \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &= \frac{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}{\pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q})} \\ &= \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})} ((p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \leq 0 \end{aligned}$$

705 The last inequality uses the case-2 condition together with  $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$ , which implies  
 706  $p^*(\mathbf{r}_i) < \text{ACC}_{\theta(t)}$  for every non-optimal pattern  $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*\}$  and all  $t \geq 0$ . Since this gives  
 707  $\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq 0$ , the ratio comparison yields:

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) = \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i)} \geq \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i)}$$

708 While  $\text{ACC}_{\theta(t)} < p^*(\mathbf{r}')$ , define  $\gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})}$ . This reference-probability ratio  
 709 controls the margin in the differential inequality:

$$\begin{aligned} \frac{d}{dt} \rho_{\theta(t)}(\mathbf{r}_i) &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\gamma_{\pi_{\text{ref}}}} \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}} \end{aligned}$$

710 Therefore,

$$\frac{d}{dt} \rho_{\theta(t)}(\mathbf{r}_i) \leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}, \quad (\text{C.7})$$

711 and this differential inequality implies:

$$\rho_{\theta(t)}(\mathbf{r}_i) \leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2t)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \rho_{\theta(0)}(\mathbf{r}_i).$$

712 Here  $\Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}')$ . To compare the aggregate accuracy with the suboptimal pattern's success  
 713 rate, recall that:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &= \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r}) - p^*(\mathbf{r}') \\ &= p^*(\mathbf{r}^*) \left[ (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) + \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}_i) - p^*(\mathbf{r}')) \right] \\ &\geq p^*(\mathbf{r}^*) \left[ (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right]. \end{aligned}$$

714 Let  $C_2 = 1/\Delta$  and  $C_1 = p^*(\mathbf{r}')/\Delta$ , and define the transition time:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left( (C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right),$$

715 At this time scale, the accumulated probability ratio is bounded as follows:

$$\begin{aligned} &\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(T_0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2T_0)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{\left( 1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) \cdot \left( \frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) \\ &= \frac{1}{\left( \left( \frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \gamma_{\pi_{\text{ref}}} \\ &= \Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}'). \end{aligned}$$

716 Finally, for any  $t > T_0$ , the previous bound makes the accuracy margin nonnegative:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &\geq p^*(\mathbf{r}^*) \left[ (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right] \\ &\geq p^*(\mathbf{r}^*) [(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - (p^*(\mathbf{r}^*) - p^*(\mathbf{r}'))] = 0. \end{aligned}$$

717 This concludes the proof of the theorem.  $\square$

718 **C.5 Proof of Theorem 5.6**

719 *Proof of Theorem 5.6.* First, consider the binary-answer case  $|\mathcal{A}| = 2$  with the uniform base dis-  
 720 tribution  $\pi_{\theta_{\text{base}}}(\mathbf{r}|q) = \frac{1}{|\mathcal{R}|}$ . Under Assumption 5.5, the average pattern success rate satisfies  
 721  $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) > 0.5$ . The derivative of aggregate accuracy can then be written as

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|q) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i|q) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \left( - \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a}|\mathbf{r}_i, q)) - \text{Con}_{\theta(t)} \right), \end{aligned}$$

722 where  $\text{Con}_{\theta(t)} = - \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}_i|q) \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a}|\mathbf{r}_i, q))$ . Let  $\mathcal{A} = \{\mathbf{a}^*, \mathbf{a}'\}$ , where  $\mathbf{a}'$  is  
 723 the incorrect answer, and initialize  $\pi_{\theta(0)}(\mathbf{r}_i|q) = \frac{1}{|\mathcal{R}|}$  for all  $\mathbf{r}_i \in \mathcal{R}$ . The initialization derivative  
 724 becomes:

$$\begin{aligned} \left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} &= \sum_{\mathbf{r}_i \in \mathcal{R}} \frac{1}{|\mathcal{R}|^2} \left( p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \\ &\quad \left( - \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a}|\mathbf{r}_i, q)) + \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a}|\mathbf{r}_j, q)) \right) \\ &= \frac{1}{|\mathcal{R}|^2} \sum_{\mathbf{r}_i \in \mathcal{R}} \left( p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \left( - \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a}|\mathbf{r}_i, q)) \right). \end{aligned}$$

725 Now take the limit  $|\mathcal{R}| \rightarrow +\infty$  with success rates  $p^*(\mathbf{r})$  drawn i.i.d. from  $U[0, 1]$ . By the Central  
 726 Limit Theorem, for large  $|\mathcal{R}|$ , the sample mean  $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r}))$  concentrates around its expectation  
 727 0.5 with deviation order  $\mathcal{O}(1/\sqrt{|\mathcal{R}|})$ . We denote this positive deviation by  $\delta = \mathcal{O}(1/|\mathcal{R}|^{1/2}) > 0$ ,  
 728 so  $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) = 0.5 + \delta$ .

729 By the Gibbs conditioning principle ([12] Corollary 7.3.5 and Theorem 7.3.8), as  $|\mathcal{R}| \rightarrow \infty$ , the  
 730 empirical distribution of  $p^*(\mathbf{r})$  conditioned on the mean  $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) = 0.5 + \delta$  converges  
 731 to the density  $f(x) = 1 + 12\delta(x - 0.5)$ ,  $x \in [0, 1]$ . We can therefore replace the summation over  
 732 reasoning patterns with an integral over this limiting distribution, giving the initialization derivative:

$$\left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} = \int_0^1 - (1 + 12\delta(x - 0.5)) \cdot (x - 0.5 - \delta) \cdot \log(x(1 - x)) dx = \frac{2}{3}\delta > 0$$

733 Since  $\delta > 0$ , the derivative is positive. As  $|\mathcal{R}| \rightarrow \infty$ , this holds with probability 1, proving the first  
 734 statement.

For the second statement, we analyze how RLIF ranks paths when  $|\mathcal{A}| = 2$ . The path reward is proportional to the negative log-likelihood of the policy:

$$R_{\text{RLIF}}(\mathbf{r}) - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|q, \mathbf{r})) = -\frac{1}{2} \log(p^*(\mathbf{r})(1 - p^*(\mathbf{r})))$$

735 The function  $g(p) = p(1 - p)$  is maximized at  $p = 0.5$ . Therefore, maximizing the reward  $R_{\text{RLIF}}(\mathbf{r})$   
 736 is equivalent to choosing the path whose success rate  $p^*(\mathbf{r})$  is furthest from 0.5.

Let  $p_{\min}^* = \min_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$  and  $p_{\max}^* = \max_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ . The path with the highest reward will be the one with success rate  $p_{\min}^*$  if it is further from 0.5 than  $p_{\max}^*$  is. This condition is expressed as:

$$0.5 - p_{\min}^* > p_{\max}^* - 0.5 \iff p_{\min}^* + p_{\max}^* < 1$$

737 Thus, the event  $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$  is equivalent to the event  $p_{\min}^* + p_{\max}^* < 1$ .

738 We now calculate the probability of this event. The success rates  $p^*(r)$  are drawn from the limiting  
739 distribution  $f(x) = 1 + 12\delta(x - 0.5)$ . From the theory of order statistics, the scaled minimum and  
740 maximum of  $n = |\mathcal{R}|$  samples converge in distribution to independent exponential random variables.  
741 Let  $Y_n = n \cdot p_{\min}^*$  and  $Z_n = n \cdot (1 - p_{\max}^*)$ . As  $n \rightarrow \infty$ :

- 742 •  $Y_n$  converges to  $Y \sim \text{Exp}(\lambda_Y)$ , where the rate is the density at the lower bound:  $\lambda_Y = f(0) =$   
743  $1 - 6\delta$ .
- 744 •  $Z_n$  converges to  $Z \sim \text{Exp}(\lambda_Z)$ , where the rate is the density at the upper bound:  $\lambda_Z = f(1) =$   
745  $1 + 6\delta$ .

The condition  $p_{\min}^* + p_{\max}^* < 1$  can be rewritten in terms of these asymptotic variables:

$$\frac{Y_n}{n} + \left(1 - \frac{Z_n}{n}\right) < 1 \implies Y_n < Z_n$$

The probability of this event is  $P(Y < Z)$ . For two independent exponential variables, this is given by:

$$P(Y < Z) = \frac{\lambda_Y}{\lambda_Y + \lambda_Z} = \frac{1 - 6\delta}{(1 - 6\delta) + (1 + 6\delta)} = \frac{1 - 6\delta}{2} = \frac{1}{2} - 3\delta$$

Since  $\delta = O(1/|\mathcal{R}|^{1/2})$ , in the limit as  $|\mathcal{R}| \rightarrow \infty$ ,  $\delta \rightarrow 0$ . Therefore, the limiting probability is:

$$\lim_{|\mathcal{R}| \rightarrow \infty} P(\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})) = \lim_{\delta \rightarrow 0} \left(\frac{1}{2} - 3\delta\right) = \frac{1}{2}$$

746 This establishes the stated 50% limiting probability and completes the proof of the second statement.

747 □

## 748 **D Full Prompts**

### 749 **D.1 Full Prompt for Reasoning Patterns Extraction**

#### Full Prompt for Reasoning Patterns Extraction

You will analyze multiple AI reasoning processes, showing how different models solve problems.

Analysis Steps:

For each reasoning process, identify:

Key words and recurring phrases

Logical structure of the argument

Problem-solving techniques used

Step-by-step progression

Group similar reasoning processes into exactly 5 patterns based on shared:

Vocabulary patterns (common terms and phrases)

Logical frameworks (how arguments are structured)

Solution approaches (methods used to reach conclusions)

**IMPORTANT!** You need to categorize general thinking patterns for problem solving, rather than categorizing types of problems themselves. In other words, each type of thinking pattern you give can be applied to any problem, rather than being task-specific. What you need to classify is the way of thinking, not the way of solving problems.

For example, you can classify them into:

1. derivation based on existing knowledge

2. derivation combined with verification

3. independent calculation, suitable for learners starting from the basics,

4. detailed manual derivation, suitable for learners who need to deeply understand each step

5. etc

Or you can classify them into:

1. knowledge call type

2. derivation and verification type

3. etc.

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Output Format:  
 Pattern 1:  
 Name and description  
 Defining characteristics  
 Example processes from input  
 Common elements  
 Pattern 2:  
 Name and description  
 Defining characteristics  
 Example processes from input  
 Common elements  
 ...  
 There are reasoning processes:  
 {extracted\_responses}  
 Group similar reasoning processes into exactly 5 patterns. Please provide a classification of reasoning patterns, with each group defined by their common characteristics and examples from these processes.

751

## 752 D.2 Full Prompt for Responses Classification

### Full Prompt for Responses Classification

Please classify each reasoning process into one of the following five patterns. Output the classifications in JSON format.  
 {patterns\_categories}  
 Here are the reasoning processes to classify:  
 {extracted\_responses}  
 Please output your classification in this JSON format:

753

## 754 D.3 Prompt Examples for Evaluation Tasks

### Prompt Example for Easy Countdown

<lim\_start>system  
 You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim\_end>  
 <lim\_start>  
 user Using the numbers [1,2,3,7], create an equation that equals 5. You can only use basic arithmetic addition (+) and subtraction (-), and each number from the list can be used at most once. Show your work in <think> </think> tags. And return the final answer in <answer> </answer> tags, for example, to get 5 from [1,2,3,7], we have <answer> 2 + 3 = 5 </answer>; to get 13 from [1,2,3,7], we have <answer> 1 + 2 + 3 + 7 = 13 </answer>.<lim\_end>  
 <lim\_start>assistant  
 Let me solve this step by step.  
 <think>

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### Prompt Example for Long Arithmetic

<lim\_start>system  
 You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim\_end>  
 <lim\_start>user  
 Considering the experssion (((((((((16 + 47) + 92) + 84) + 5) + 49) + 82) - 33) - 99) - 69), you need to start from the innermost bracket and work your way out. You need to give the specific

756

calculation process. You have to do a certain amount of calculations on each line, and print each calculation in the format of  $a + b = c$  or  $a - b = c$ , for example " $1 + 2 = 3$ ".  
 <lim\_start>assistant  
 Let me solve this step by step.  
 <think>

757

#### Prompt Example for Number Theory

<lim\_start>system  
 You are a helpful assistant.<lim\_end>  
 <lim\_start>user  
 You have four bags of copper coins. Each bag has the same number of copper coins. One day, you find a bag of 23 coins. You decide to redistribute the number of coins you have so that all five bags you hold have the same number of coins. You successfully manage to redistribute all the coins, and you also note that you have more than 120 coins. What is the smallest number of coins you could have had before finding the bag of 23 coins?  
 Please reason step by step, and put your final answer within  $\boxed{\phantom{000}}$ .<lim\_end>  
 <lim\_start>assistant

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#### Prompt Example for Geometry

<lim\_start>system  
 You are a helpful assistant.<lim\_end>  
 <lim\_start>user  
 Two sides of scalene  $\triangle ABC$  measure 4 centimeters and 7 centimeters. How many different whole centimeter lengths are possible for the third side?  
 Please reason step by step, and put your final answer within  $\boxed{\phantom{000}}$ .<lim\_end>  
 <lim\_start>assistant

759

#### Prompt Example for Algebra

<lim\_start>system  
 You are a helpful assistant.<lim\_end>  
 <lim\_start>user  
 Six aluminum cans can be recycled to make a new can. How many new cans can eventually be made from 200 aluminum cans? (Remember that the first new cans that are made can then be recycled into even newer cans!)  
 Please reason step by step, and put your final answer within  $\boxed{\phantom{000}}$ .<lim\_end>  
 <lim\_start>assistant

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#### Prompt Example for Calculus

<lim\_start>system  
 You are a helpful assistant.<lim\_end>  
 <lim\_start>user  
 For how many values of  $x$  in  $[0, 2\pi]$  is  $\cos^{-1}(\cos 4x) = \sin^{-1}(\sin x)$ ?  
 Please reason step by step, and put your final answer within  $\boxed{\phantom{000}}$ .<lim\_end>  
 <lim\_start>assistant

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#### Prompt Example for Counting and Probability

<lim\_start>system  
 You are a helpful assistant.<lim\_end>  
 <lim\_start>user

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7 white balls and  $k$  black balls are placed into a bin. Two balls are drawn at random. The probability that one ball is white and the other is black is  $\frac{35}{66}$ . Find the smallest possible value of  $k$ .  
Please reason step by step, and put your final answer within  $\boxed{\phantom{000}}$ .<lim\_endl>  
<lim\_startl>assistant

763

#### Prompt Example for AMC23-Q19

<lim\_startl>system  
You are a helpful assistant.<lim\_endl>  
<lim\_startl>user  
How many digits are in the base-ten representation of  $8^5 \cdot 5^{10} \cdot 15^5$ ?  
Please reason step by step, and put your final answer within  $\boxed{\phantom{000}}$ .<lim\_endl>  
<lim\_startl>assistant

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#### Prompt Example for AIME24-Q1

<lim\_startl>system  
You are a helpful assistant.<lim\_endl>  
<lim\_startl>user  
Every morning Aya goes for a 9-kilometer-long walk and stops at a coffee shop afterwards. When she walks at a constant speed of  $s$  kilometers per hour, the walk takes her 4 hours, including  $t$  minutes spent in the coffee shop. When she walks  $s+2$  kilometers per hour, the walk takes her 2 hours and 24 minutes, including  $t$  minutes spent in the coffee shop. Suppose Aya walks at  $s+\frac{1}{2}$  kilometers per hour. Find the number of minutes the walk takes her, including the  $t$  minutes spent in the coffee shop.  
Please reason step by step, and put your final answer within  $\boxed{\phantom{000}}$ .<lim\_endl>  
<lim\_startl>assistant

765

#### Prompt Example for OlympiadBench-Q1631

<lim\_startl>system  
You are a helpful assistant.<lim\_endl>  
<lim\_startl>user  
For a positive integer  $a$ , define a sequence of integers  $x_1, x_2, \dots$  by letting  $x_1 = a$  and  $x_{n+1} = 2x_n + 1$  for  $n \geq 1$ . Let  $y_n = 2^{x_n} - 1$ . Determine the largest possible  $k$  such that, for some positive integer  $a$ , the numbers  $y_1, \dots, y_k$  are all prime.  
Please reason step by step, and put your final answer within  $\boxed{\phantom{000}}$ .<lim\_endl>  
<lim\_startl>assistant

766